

Client Pitch

Team: *Big Red Stompers* **Clients:** Cornell CALS Extension / E&J Gallo Winery / National Grape

Problem Statement

Spotted lanternflies (SLF) mainly target grapevines. As a result, grape farmers are trying to manage SLF populations in vineyards and nearby trees. SLF egg masses contain around 30-60 eggs, creating rapid population growth that affects later growing seasons. Current SLF population reduction methods, such as netting and manual egg removal are difficult to scale across larger vineyards.

Impact

Elimination of SLF eggs prevents the development of adult populations causing the most damage to vineyards. The average SLFs per vine starts low, increasing sharply following the transition from nymph to adult stages (Figure 1). As a result, attacking the root cause would protect regions like Lake Erie and the Finger Lakes from losses between \$1.5 million in year one and \$8.8 million by year three of infestation (Hayes).

Proposed Concept: Stomp n' Scrape

What it is: Handheld egg removal device.

How it would be used: Designed to crush SLF egg mass shells and immediately scrape them into a rubbing-alcohol-filled vessel to prevent hatching.

Why it's better than the status quo:

- Reduces the chance of leaving viable eggs behind.
- Minimizes mess and user discomfort by sealing egg material instantly.
- Improved output and efficiency compared to manual scraping alone.

End-of-semester proof-of-concept: A functional prototype with a crushing feature, integrated scraper, and removable alcohol container, fabricated using 3D-printed components and accessible materials, and tested on simulated SLF egg masses to evaluate effectiveness and ease of cleaning.

Key Risks / Unknowns

- 1 - Egg masses harden, making it difficult to ensure they are fully damaged. Two methods to kill the egg masses: crushing them and then scraping them into rubbing alcohol to kill the remaining ones.
- 2 — Public aversion to kill/interact with SLF, as crushing egg masses may be off-putting. Test with user trials and gauge user comfort.
- 3 — Many eggs are laid on high surfaces; 98% of eggs are found at unreachable heights (+10 feet) on trees (Krawczyk). Test attachments to extendable tools for users to reach higher egg masses.

Questions for Client

1. **What is preventing you from currently targeting egg masses? Is it location, time or resources?**

Decision affected: This would affect the aspects we want to emphasize in our design, like making it accessible to higher areas of the tree and ease to clean, based on client demand.

2. **Are there methods for reaching high areas that you are currently using?**

Decision affected: This would affect whether our product must account for reaching high-up areas, or if it can be used with existing tools.

3. **How much would you need to reduce SLF at the egg stage to be considered effective?**

Decision affected: This would affect how we determine the force needed and how we design our crushing/scraping mechanism.

References

- New York State Integrated Pest Management. (n.d.). Spotted lanternfly management. <https://cals.cornell.edu/integrated-pest-management/outreach-education/whats-bugging-you/spotted-lanternfly>
- Hayes, C. (2025, January 27). Spotted lanternflies could cost NYS grape industry millions. Cornell Chronicle. <https://news.cornell.edu/stories/2025/01/spotted-lanternflies-could-cost-nys-grape-industry-millions#:~:text=Using%20data%20from%20two%20key,and%20third%20years%20of%20infestation%2C>.
- Krawczyk, G. (2023, August 30). What should you do with spotted lanternfly egg masses? PennState Extension. <https://extension.psu.edu/what-should-you-do-with-spotted-lanternfly-egg-masses>

Figure

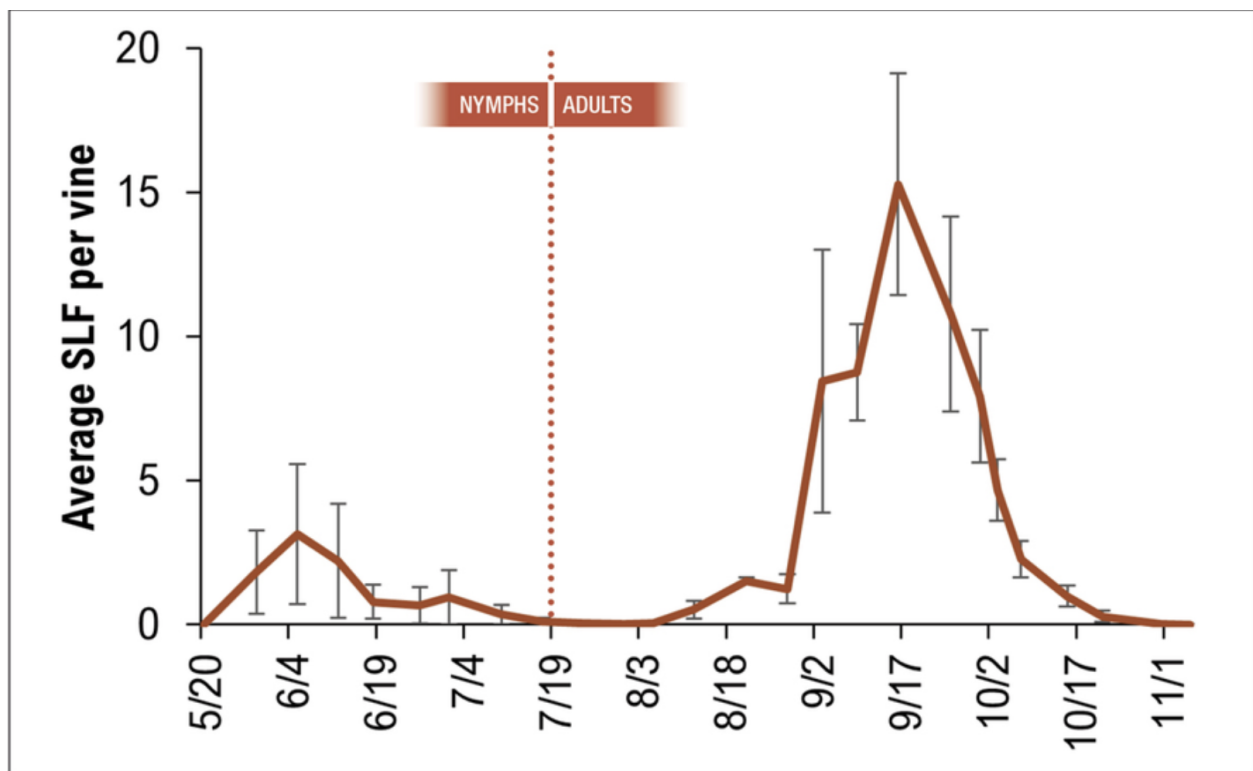


Figure 1: Average SLFs per Vine (2018-2020)